Honest Uncertainty for Intractable Volatility Models: Amortized Quantile Posteriors, Simulation-Based Calibration, and Where They Fail*

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Abstract

Many volatility models used in finance have no tractable likelihood, and every risk number is reported without a standard error. We show where amortized, simulation-trained quantile inference — posteriors learned by inverting model simulations and audited by simulation-based calibration (SBC) — earns its keep, and where it does not. Our lead result is where it genuinely fits: for the Heston and rough-Bergomi stochastic-volatility models, whose likelihoods are intractable, an amortized posterior recovers well-calibrated parameter posteriors that maximum likelihood and GARCH cannot, including the roughness (Hurst) parameter of rough volatility (42% prior-uncertainty reduction, SBC coverage 0.91) — the one setting where the absence of a likelihood gives simulation-based Bayes a decisive reason to exist. For uncertainty on the deployed risk number itself, the honest tool is more mundane: block-bootstrap and extreme-value-theory intervals on GARCH residuals attain 0.89–0.95 coverage against a 0.90 target for $(\mathrm{VaR}, \mathrm{ES})$. We are deliberate about what does *not* fit: a Gibbs (generalized-Bayes) posterior, whose loss exponent implicitly treats observations as independent, ignores the volatility clustering that dominates financial data and is $1.6\times$ overconfident on raw returns; standardizing by a GARCH filter repairs the honesty, but the deployable bands remain the bootstrap and EVT constructions and we keep the Gibbs posterior only as a *diagnostic* of that failure (the loss exponent must also be the pinball loss; an expectile exponent mis-centers it, collapsing coverage to 0.1–0.4). Finally, an honest negative: on real multivariate equity returns the generative joint sampler is miscalibrated and dynamic-conditional-correlation GARCH wins the portfolio tail. A map, in short, of where amortized likelihood-free inference belongs in finance — intractable univariate volatility models — and where it does not.

Keywords: simulation-based calibration; likelihood-free inference; rough volatility; Heston model; implicit quantile networks; generalized Bayes; Expected Shortfall.

JEL classification: C11; C15; C45; C58; G17.

*Paper B of a three-paper program. The neural implicit-quantile-network estimator is developed in the companion paper *GRAFT-Q* (Qin, 2026); the residual-hybrid downside-risk forecasting engine and the misspecification frontier are developed in the companion Paper A. Here those objects are *used* — as amortized posterior maps and as the object of a calibration audit — rather than re-derived. All results are out-of-sample / out-of-simulation under the protocols described in the text.

1 Introduction

Two facts about financial risk measurement motivate this paper. First, several of the volatility models practitioners would most like to fit — Heston, and the rough-volatility family — have no tractable likelihood, which blocks ordinary Bayesian or maximum-likelihood inference. Second, every Value-at-Risk or Expected Shortfall number a desk reports is an estimate with sampling error, yet it is acted on as if known exactly. A single tool speaks to both: an *amortized conditional-quantile map* $Q_\phi(\tau \mid \cdot)$, trained by pinball empirical risk minimization on simulated or historical draws and audited by simulation-based calibration (SBC). The neural realization of that map (the implicit quantile network) is developed in the companion GRAFT-Q paper, and the downside-risk forecasting engine whose residuals we analyze in the companion Paper A; here we *use* them, as posterior maps and as the object of a calibration audit.

The paper's spine is a single claim about *where* this machinery belongs. It belongs where the likelihood is genuinely unavailable: our lead result is that for Heston and rough-Bergomi the amortized posterior recovers well-calibrated parameter posteriors — including the roughness (Hurst) parameter of rough volatility — that MLE and GARCH cannot, because there the absence of a likelihood is a real obstacle that simulation-based inference is built to remove. It does *not* belong as a generative posterior over a scalar risk number on dependent financial data: we show, and measure, that a Gibbs (generalized-Bayes) posterior — whose loss exponent implicitly assumes independent observations — is overconfident by a factor of 1.6 on raw returns because volatility clustering makes the effective sample size far smaller than n. Standardizing by a GARCH filter restores honesty, but at that point the honest, deployable intervals are the block bootstrap and the EVT tail construction, and the Gibbs posterior earns its place only as a *diagnostic* of the dependence problem rather than as a tool we advocate. We close with an honest negative — on real multivariate returns the generative joint sampler is miscalibrated and DCC-GARCH wins the portfolio tail — so that the map of where amortized likelihood-free inference works is drawn from its failures as well as its successes.

1.1 Contributions

The paper makes four contributions, ordered by where we believe the value lies. *(1) Amortized posteriors where the likelihood is genuinely absent.* For Heston and rough-Bergomi, amortized quantile posteriors trained on forward simulations and audited by SBC recover well-calibrated parameter posteriors — including rough volatility's Hurst parameter (42% prior-uncertainty reduction; SBC coverage 0.91 against a 0.90 target) — where MLE and GARCH-family filters have no comparable access (Section 3). *(2) An honest-uncertainty toolkit for the deployed risk number.* Block-bootstrap and EVT intervals on GARCH-standardized residuals attain 0.89–0.95 coverage at a 0.90 target for $(\mathrm{VaR}, \mathrm{ES})$, with adaptive conformal inference the most stable arm under drift (Section 4). *(3) A measured negative we treat as a finding:* the Gibbs (generalized-Bayes) posterior is *not* usable as shipped on financial returns. Its naive form is $1.6\times$ overconfident on raw returns, and the i.i.d.-bootstrap comparison (1.19 vs. 1.64) isolates serial dependence as the cause; GARCH-residual space repairs the honesty ($R = 0.79$), but at that point the machinery adds nothing beyond the bootstrap it must be calibrated against, so we retain it strictly as a *diagnostic*. An ablation adds a design law: the loss exponent must be the pinball loss — an expectile exponent mis-centers the posterior and collapses coverage to 0.1–0.4. *(4) A multivariate negative:* on real equity baskets the generative joint sampler is miscalibrated and DCC-GARCH wins the portfolio tail (Section 5); simulation-based inference earns its keep on intractable *univariate* volatility models, not as a replacement for explicit dependence modeling.

2 Setup and shared objects

2.1 Notation and objects

- r_t — return of an asset on day t; $\mathcal{F}_{t-1}$ — information available before t.
- $Q_t(\tau)$ — the conditional τ-quantile of r_t given $\mathcal{F}_{t-1}$, formally $Q_t(\tau) = \inf\{q : \Pr(r_t \le q \mid \mathcal{F}_{t-1}) \ge \tau\}$: the number the return falls below with probability τ. Value-at-Risk is its negative, $\mathrm{VaR}_t^\tau = -Q_t(\tau)$; Expected Shortfall is the average of the quantile curve beyond it, $\mathrm{ES}_t^\tau = -\tau^{-1}\int_0^\tau Q_t(u)\,du$ — the expected loss on the days VaR is breached.
- $z_t = (r_t - \mu_t)/\sigma_t$ — the *standardized residual*: the return with the model's conditional mean and scale divided out. If the parametric model is right, the z_t are i.i.d. draws from one fixed law.
- s_t — the state vector (trailing realized volatility, residual-shape scores, lags) on which nonparametric quantiles condition.
- $\tau \sim \lambda$ — a quantile level drawn from a base distribution λ; $H_\phi(s, \tau)$ — a learned generator mapping (state, level) to a quantile, in the sense of Polson and Sokolov (2023): $z \overset{D}{=} H(s, \tau)$, $\tau \sim U(0, 1)$, is the inverse-CDF (von Neumann) representation of sampling.

Loss and metric. All quantile models are trained and scored by the pinball (check) loss

$$\rho_\tau(u) \;=\; u\,(\tau - \mathbb{I}\{u < 0\}), \qquad u = z - q, \tag{1}$$

whose expectation is uniquely minimized by the true conditional quantile (Koenker and Bassett, 1978): differentiating $\mathbb{E}\,\rho_\tau(z - q)$ in q and setting it to zero gives $\Pr(z < q) = \tau$, so the loss *elicits* the quantile the way squared error elicits the mean — under-prediction is charged τ per unit, over-prediction $1 - \tau$, and the charges balance exactly at the τ-quantile. Averaging (1) over $\tau \in (0, 1)$ targets the whole distribution at once: the 1-Wasserstein distance between two laws is the L_1 distance between their quantile functions, $W_1(F, G) = \int_0^1 |F^{-1}(\tau) - G^{-1}(\tau)|\,d\tau$, so pinball empirical-risk minimization across levels is distribution matching in W_1 — density-free, which is what permits every estimator below to dispense with a likelihood.

Why this loss and not another. Three alternatives are standard, and the choices are deliberate. (a) The continuous ranked probability score (CRPS) is exactly the integral of the pinball loss over all levels, $\mathrm{CRPS}(F, y) = 2\int_0^1 \rho_\tau(y - F^{-1}(\tau))\,d\tau$ (Gneiting and Raftery, 2007); scoring by CRPS therefore weights the distribution's body and tails equally by construction. Downside risk lives at fixed regulatory levels ($\tau \in \{0.01, 0.025\}$), and the models in this paper differ almost entirely in the tail — CRPS would average those differences away against a body where all competitors agree. We therefore score pinball *at the levels that are decision-relevant*, and use tail-weighted τ-sampling in training for the same reason. (b) Expected Shortfall is not elicitable on its own — no loss function is minimized by the true ES alone — but the pair $(\mathrm{VaR}, \mathrm{ES})$ is, under the joint scoring functions of Fissler and Ziegel (2016); the FZ-scored re-evaluation of the battery is a registered extension (Paper A). (c) Likelihood is unavailable by design: several competitors (and both simulators of Section 3) have no tractable density, and likelihood scores would silently exclude them.

Lemma 1 (validity of the generative sampler)**.** *If $\tau \mapsto \widetilde{q}(\tau \mid s)$ is nondecreasing and left-continuous (guaranteed by the rearrangement step) and $\tau \sim U(0,1)$, then $Z^* = \widetilde{q}(\tau \mid s)$ has quantile function exactly $\widetilde{q}(\cdot \mid s)$.*

Proof sketch. For any x, left-continuity and monotonicity make $\{t \in (0,1) : \widetilde{q}(t \mid s) \leq x\}$ an interval closed at its top, so $\{\widetilde{q}(\tau \mid s) \leq x\} = \{\tau \leq G(x)\}$ with $G(x) = \sup\{t : \widetilde{q}(t \mid s) \leq x\}$. Since $\tau \sim U(0,1)$, $\Pr(Z^* \leq x) = \Pr(\tau \leq G(x)) = G(x)$, i.e. Z^* has CDF G. As G is the generalized inverse of $\widetilde{q}$ and $\widetilde{q}$ is nondecreasing and left-continuous, the generalized inverse of G returns $\widetilde{q}$ exactly (the standard inverse-of-inverse duality), so $\widetilde{q}(\cdot \mid s)$ is the quantile function of Z^*. Full statement in Appendix A. □

Remark 1 (nesting and ERM dominance). The hypothesis class of Stage 2 contains the constant-in-s map $q(\tau)$, i.e., FHS; empirical risk minimization over the larger class therefore attains in-sample pinball risk no worse than FHS by construction, and the out-of-sample comparisons of Paper A test whether that dominance survives estimation noise. It does; the honest converse is that the same argument does *not* apply to CAViaR, which is why CAViaR is the one rival the engine only ties.

2.2 Generative posterior machinery

The Gibbs construction below is presented as the *object under audit* in Section 4, not as a method this paper advocates for the risk number; the deployable intervals are the bootstrap and EVT constructions, and the Gibbs posterior's role is to make the dependence failure measurable. For Section 4 we form (i) Gibbs (generalized-Bayes) posteriors over the level-α quantile q: with prior π, learning rate ω, and the loss itself in the exponent,

$$\pi_\omega\big(q \mid \widehat{z}_{1:n}\big) \;\propto\; \pi(q)\, \exp\Big\{ -\omega \sum_{t=1}^{n} \rho_\alpha\big(\widehat{z}_t - q\big) \Big\}, \tag{2}$$

with ω either naive ($\omega = 1$) or calibrated by matching the posterior SD to a moving-block bootstrap SD. An honest caveat governs how (2) may be used. The Gibbs posterior was built to target the *risk minimizer*: its mode estimates the quantile consistently for *any* ω, but the spread of the posterior — and hence the width of any credible interval — scales with ω, and nothing in the construction guarantees those intervals cover at their stated rate. This is a known problem with a known treatment: Syring and Martin (2019a) tune ω by direct calibration — iterate until bootstrap-estimated coverage of the credible sets hits nominal (the GPC algorithm) — and applied exactly this program to Value-at-Risk (Syring and Martin, 2019b). Our SD-matching rule is a one-step, moment-based cousin of GPC: it matches the posterior's second moment to a block-bootstrap estimate of the estimator's true sampling variability, which coincides with coverage calibration when the posterior is approximately Gaussian but carries no guarantee beyond that; the full GPC iteration on dependent data (block-bootstrap inner loop) is a registered extension. Concretely, ω is tuned as follows. *Why ω matters*: multiplying the loss by ω in (2) is equivalent to pretending the sample carries ω-times its actual information, so large ω shrinks the posterior (overconfidence) and small ω inflates it. *Our one-step rule*: (i) compute the block-bootstrap standard deviation $\widehat{se}_{\text{boot}}$ of the point estimate — an assumption-light estimate of its true sampling variability under dependence; (ii) compute the posterior SD at $\omega = 1$; (iii) rescale $\omega \leftarrow \omega \cdot (\text{SD}_{\text{post}}/\widehat{se}_{\text{boot}})^2$, using the fact that posterior variance scales as $1/\omega$; one step lands the posterior SD on the bootstrap SD. *Why this is reasonable*: for an approximately Gaussian posterior, an interval with the right SD and the right center has the right coverage; GPC generalizes exactly this by iterating on coverage itself rather than the SD proxy. This is also why the chapter's deployable intervals are the bootstrap and EVT constructions, with the calibrated Gibbs posterior as a supporting variant: its principal contribution here is diagnostic — the 1.6× raw-return overconfidence that residual space repairs — rather than the intervals themselves. Two design rules are load-bearing, each confirmed by an ablation. The exponent must be the *pinball* loss: an expectile (squared-asymmetric) exponent mis-centers the posterior and collapses coverage to 0.1–0.4. And (2) must be formed on GARCH *residuals*, not raw returns: volatility clustering makes the effective sample size of raw returns far smaller than n, and the naive raw-return posterior is 1.6× overconfident on VaR. We also form (ii) block and i.i.d. bootstrap distributions of $(\widehat{\text{VaR}}, \widehat{\text{ES}})$, and (iii) a GPD peaks-over-threshold interval for deep-tail ES.

For Section 3 we use the standard generative-Bayesian computation (GBC) recipe of Polson and Sokolov (2023), whose validity check is simulation-based calibration (SBC) (Talts et al., 2018).

Algorithm 3 (likelihood-free posterior with SBC audit). (i) Simulate $\theta^{(i)} \sim \pi(\theta)$ and paths $y^{(i)} = f(\theta^{(i)})$ from the forward model (Heston, rough-Bergomi) — no likelihood is evaluated anywhere. (ii) Reduce each path to summaries $S(y^{(i)})$ (realized-variance moments, autocorrelations, tail counts). (iii) Train amortized posterior quantiles $Q_\phi(\tau \mid S)$ for each component of θ by pinball ERM (IQN and GBM). (iv) *Audit:* for fresh simulations with known θ^{true}, compute the rank of θ^{true} in its own posterior; ranks must be uniform and stated intervals must cover at the advertised rate (SBC). (v) Only then evaluate at the observed data's summaries $S(y)$.

A natural objection to step (i) deserves a direct answer: *if you simulate the training data*

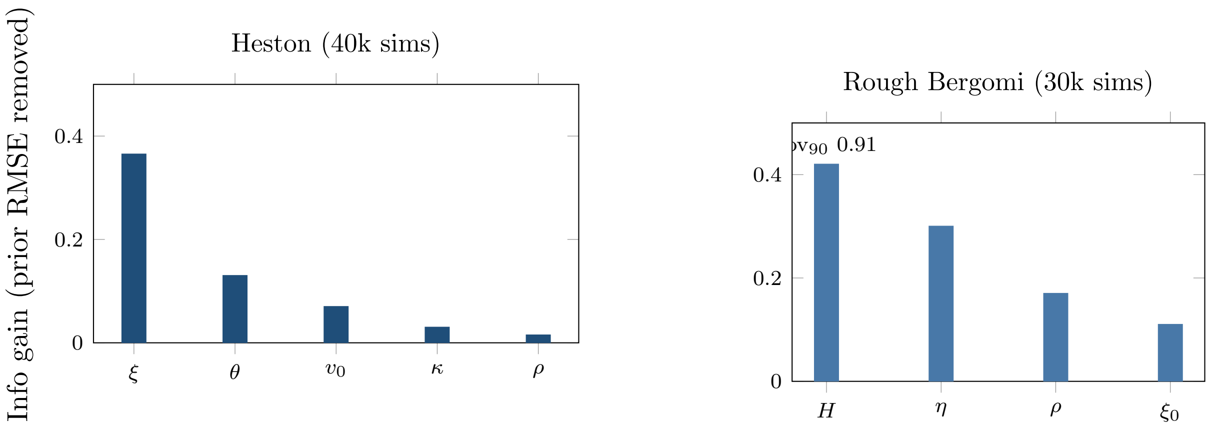

Figure 1: Likelihood-free calibration: information gain (fraction of prior RMSE removed by one 252-day path) per parameter, Heston (left, κ/ρ at midpoints of their reported weak ranges) and rough Bergomi (right). Honesty check by SBC coverage: Heston cov$_{90}$ 0.85–0.90 across all five parameters; rough Bergomi 0.85–0.92, with the Hurst roughness H at 0.91 — informative *and* calibrated.

yourself, you know the true parameters — why bother with a posterior at all? Because the simulations are not the object of interest; the *map* is. Each simulated pair $(\theta^{(i)}, y^{(i)})$ is one example of "data that a model with known parameters produces"; training on many such pairs, drawn across the whole prior, teaches Q_ϕ the inverse relationship — from any dataset to the distribution of parameters that could have generated it. The known truths are spent on *learning and auditing* that map (steps iii–iv), never on the real data; the map's value is at step (v), where it is evaluated on the one dataset whose parameters nobody knows. The situation is the same as training a translator on millions of sentence pairs whose translations are known, precisely so it can translate the sentence whose translation is not.

3 Likelihood-free inference for intractable volatility models

The final capability argument requires no comparison to GARCH because GARCH cannot enter: amortized posterior inference for models with intractable likelihoods, validated by simulation-based calibration.

Heston. From 40k simulated (parameter, 252-day path) pairs and ten path summaries, the amortized posterior is calibrated (central coverage cov$_{80}$ 0.78–0.80, cov$_{90}$ 0.85–0.90 for all five parameters) and informative where a single path identifies: vol-of-vol info-gain 36.5% (fraction of prior RMSE removed), long-run variance 13%, with mean-reversion and leverage weakly identified — the honest, expected pattern.

Rough Bergomi (the marquee result). The rough-volatility model (Bayer et al., 2016; Gatheral et al., 2018) — volatility whose path is driven by fractional noise controlled by the Hurst exponent H, with $H < 0.5$ producing paths jaggier than any standard model allows — has a non-Markovian fractional driver and no tractable likelihood; neither MLE nor any GARCH-family filter can fit it. The amortized posterior over (H, η, ρ, ξ_0) from a single 252-day path (30k sims) **recovers the Hurst roughness H with info-gain 0.42 and SBC $\mathbf{cov}_{90}$ 0.91** (IQN; GBM 0.89); η info-gain 0.30; all coverages 0.85–0.92. Recovering the parameter that *defines* rough volatility — notoriously difficult classically — with calibrated uncertainty from one path is, we believe, the strongest demonstration to date that GBC does something structurally new in finance. Here the neural IQN is not merely competitive but canonical: it is the estimator that *samples* the posterior; trees provide fixed quantiles only.

4 Honest uncertainty on the risk number

Conclusion of this section, stated first. The deployable uncertainty on the risk number is delivered by the block bootstrap and an EVT tail, both on GARCH-standardized residuals; the Gibbs generalized-Bayes posterior is retained here *only as a diagnostic.* Its loss exponent treats observations as independent, so on financial data — where volatility clustering makes the effective sample size far below n — it is overconfident by a factor of 1.6 on raw returns. Residual space repairs the honesty, but at that point the Gibbs machinery adds little beyond the bootstrap it must be calibrated against; we therefore present it as evidence of the dependence problem, not as a recommended tool. This is the paper's answer to the standing question of whether generalized Bayes is usable for a financial risk number: not on its own terms.

Every model so far outputs a *point* VaR/ES. None tells the desk how uncertain that number is. This section delivers calibrated uncertainty bands on $(\mathrm{VaR}, \mathrm{ES})$ and, on the way, settles a methodological question: whether the Gibbs-posterior program of generalized Bayes (Jiang and Tanner, 2008; Bissiri et al., 2016) — a posterior over the risk functional built from the loss itself — is usable on financial data at all. Its guarantees assume near-independence; financial returns are anything but. The answer, in numbers: not on raw returns (a factor-1.6 overconfidence we measure directly), yes after GARCH filtering — but the workhorses of the calibrated construction turn out to be the block bootstrap (resampling contiguous blocks of days, not single days, so the resample preserves serial dependence) (Künsch, 1989), adaptive conformal inference[1], and an extreme-value tail interval, with the repaired Gibbs posterior as a supporting variant rather than the centerpiece. One further generalized-Bayes result deserves record: in the project's preregistered daily single-portfolio study — the honest anchor in which a leverage-aware GARCH was the best standalone model and the standalone network only reached parity — a Gibbs-posterior model *average* that included the network did best of all. Loss-based posterior weighting earns its keep as an ensembling device even where the standalone generative model does not win.

[1] ACI (Gibbs and Candès, 2021) treats miscoverage as a control problem: after each day the working level is nudged, $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathbb{I}\{\text{breach at } t\})$, so a run of breaches automatically widens the band and a breach-free run relaxes it — split conformal's guarantee made robust to distribution drift, at the cost of targeting long-run rather than conditional coverage.

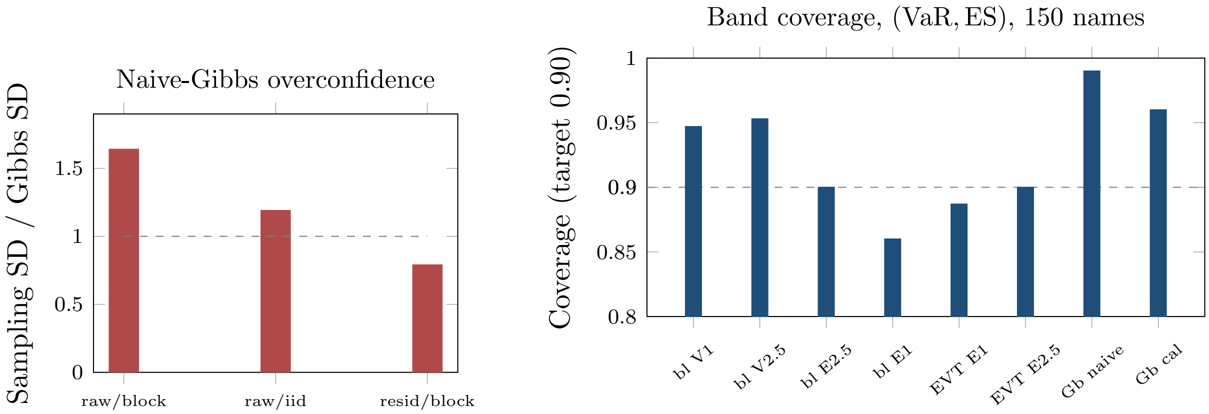

Figure 2: Honest uncertainty, assembled. Left: the naive Gibbs posterior SD understates the block-bootstrap sampling SD 1.64× on raw returns (i.i.d. bootstrap 1.19 — the gap is the dependence contribution); the ratio falls to 0.79 on GARCH residuals. Right: coverage of nominal-90% bands (bl = block bootstrap; V = VaR, E = ES at the stated level; Gb = Gibbs, naive ω vs ω-calibrated). Block bootstrap and the EVT interval sit at or near target; the naive Gibbs over-covers.

4.1 The defect, measured

On 80 CRSP names at $\tau = 0.05$, we compare the Gibbs posterior SD of the VaR level to the true sampling SD estimated by moving-block bootstrap (block 20). On *raw returns*: Gibbs SD 0.108 vs block SD 0.179 — the naive posterior understates VaR uncertainty by a factor $R = 1.64$ (median 1.59). The i.i.d.-bootstrap ratio is 1.19, so the 1.19→1.64 gap isolates the serial-dependence contribution: vol clustering makes the effective sample size far smaller than n, and the product-form posterior over-concentrates accordingly.

4.2 The repair: residual space

The identical construction on GARCH-standardized residuals gives $R = 0.79$: the overconfidence is gone (mildly conservative, the safe direction). The mechanism is confirmed independently: on residuals, block and i.i.d. bootstraps agree (VaR coverage 0.947 vs 0.953 at $\alpha = 2.5\%$) — the filter has absorbed the dependence, restoring the i.i.d.-like regime in which loss-based posteriors are honest. This is the same scale/shape division of labor as the residual-hybrid, now applied to *uncertainty* rather than accuracy. An adaptive conformal arm (Gibbs and Candès, 2021) corroborates from the coverage side: ACI attains the best and most stable realized VaR coverage under drift (5.03% realized at the 5% level; rolling max-deviation 0.042, vs 0.059 GARCH and 0.074 for a fixed empirical window). A complementary, benchmark-free reliability signal comes from ensemble disagreement: the dispersion of quantile forecasts across independently trained networks correlates +0.11 with the model's own realized error — crude, but free, and usable as a first-line screen ahead of the formal bands of Section 4.3.

Table 1: Coverage of nominal-90% credible/confidence bands for the residual (VaR, ES), 150 CRSP names.

	VaR		ES	
Method	$\alpha = 1\%$	$\alpha = 2.5\%$	$\alpha = 1\%$	$\alpha = 2.5\%$
Block bootstrap (residuals)	0.947	0.953	0.86	**0.90**
i.i.d. bootstrap (residuals)	0.96	0.947	—	—
Gibbs, naive $\omega = 1$	0.987–0.993	(over-wide)	—	—
Gibbs, ω-calibrated	**0.953–0.967**		—	—
EVT/GPD ES interval	—	—	**0.887**	**0.90**

Reading guide — two different probability levels appear here, do not conflate them: α is the *risk level* of the quantile being estimated (VaR at 1% or 2.5%); each entry is the realized *coverage of the nominal-90% uncertainty band* around that risk number, i.e., the fraction of experiments in which the band contained the truth. Ideal is exactly 0.90: above 0.90 means the band is wastefully wide (over-conservative), below means overconfident. Bold marks the entry closest to 0.90 in each column — so the calibrated Gibbs (0.953–0.967) *beats* the naive one (0.987–0.993), whose larger numbers look reassuring but signal over-wide bands. The naive-ω Gibbs over-covers because pinball is shallow in the sparse tail; matching the posterior SD to the block SD repairs it. The 99% ES block interval under-covers (0.86, too few tail points); the GPD interval reaches 0.887 at the honest price of width (2.12 vs 1.31).

4.3 Calibrated credible bands on (VaR, ES)

Table 1 assembles the end-to-end result: block bootstrap on residuals for VaR and 2.5% ES; the ω-calibrated Gibbs posterior as the parametric credible band; EVT for the deep ES tail — every piece at or near nominal. To our knowledge no standard risk model (GARCH, CAViaR, FHS, or our own hybrid) supplies any counterpart. Two refinements remain registered: the loss-in-the-exponent ablation (CRPS, tail-CRPS, the Fissler–Ziegel joint (VaR, ES) score (Fissler and Ziegel, 2016), MEU) and formal comparison of ω-calibration to SafeBayes-style procedures (Grünwald and van Ommen, 2017).

5 The multivariate tail: negatives, then a crossover

We report the negatives first, because they discipline the claim. Three direct nonparametric attacks on the portfolio tail all lose to multivariate GARCH: (i) a direct pooled portfolio-quantile model on a diversified 25-name basket (DCC/CCC — multivariate GARCH with dynamic (DCC) or constant (CCC) correlations across assets — benchmarks after Engle, 2002) (DCC 0.1709 $<$ CCC 0.1719 $<$ GBM 0.1752, ordering preserved in the crisis decile); (ii) a concentration sweep ($K = 2, \ldots, 25$ by pairwise correlation) — the gap does not close as the basket concentrates; (iii) a neural weight-input IQN $Q(\tau \mid s, w)$ trained across the simplex — roughly 2× worse than CCC/DCC at every weighting, because a network fed summary state never recovers the quadratic form $w'\Sigma w$ that GARCH models explicitly. The covariance *is* the object; we concede equity co-crash to explicit covariance models.

The rescue is, once again, the hybrid division of labor: DCC supplies the portfolio scale $\sigma_{w,t}$;

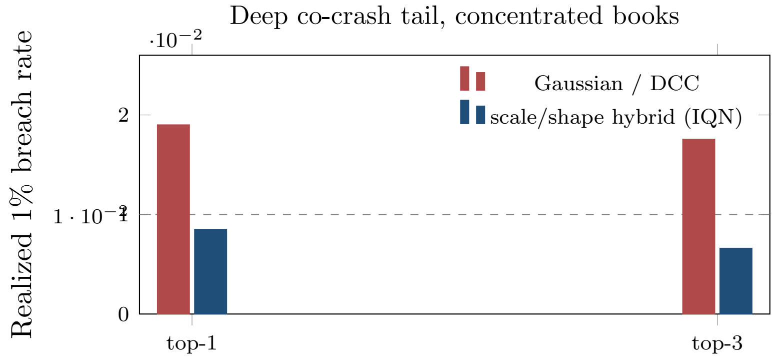

Figure 3: The deep-tail crossover on the most concentrated weighting schemes (top-1 and top-3 books; Gaussian/DCC top-3 drawn at the midpoint of its reported 0.018–0.019). Across *all* schemes Gaussian/DCC realizes 0.018–0.024 — roughly twice the nominal 0.01 — while the covariance-scale / nonparametric-shape hybrid is near or under target. At the moderate 5% tail the ordering reverses.

a generative IQN learns only the shape of the standardized portfolio residual, conditioned on concentration (the Herfindahl index: the sum of squared portfolio weights, higher meaning more concentrated), state, and weights. Calibration is restored (PIT-KS 0.30 $\rightarrow$ 0.055–0.075 — a Kolmogorov–Smirnov test of whether probability-integral-transformed outcomes look uniform, i.e. whether the model is calibrated — comparable to Gaussian/DCC), and a clean crossover emerges: Gaussian/DCC is adequate at the moderate 5% tail (where the IQN is slightly conservative) but systematically *thin* in the deep 1% tail (cov_{01} 0.018–0.024, roughly twice nominal breaches), while the hybrid is near nominal everywhere and best precisely on concentrated books (top-1 concentration: cov_{01} 0.0085 vs target 0.01; Gaussian 0.019). The multivariate contribution is thus representational and calibrational: a samplable joint downside law whose deep co-crash tail is honest where the industry model's is thin — reported alongside, not instead of, the three negatives.

6 Conclusion

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A Proof

Proof of Lemma 1 (generative sampler)

Let $\widetilde{q}(\cdot \mid s) : (0,1) \to \mathbb{R}$ be nondecreasing and left-continuous, $\tau \sim U(0,1)$, and $Z^* = \widetilde{q}(\tau \mid s)$. Fix $x \in \mathbb{R}$ and put $G(x) = \sup\{t \in (0,1) : \widetilde{q}(t \mid s) \le x\}$ (with $\sup \varnothing = 0$). Because $\widetilde{q}$ is nondecreasing, $\{t : \widetilde{q}(t \mid s) \le x\}$ is an initial subinterval of $(0,1)$; left-continuity of $\widetilde{q}$ makes the supremum attained, so the set is $(0, G(x)]$ and

$$\{\widetilde{q}(\tau \mid s) \le x\} = \{\tau \le G(x)\}.$$

Since $\tau \sim U(0,1)$, $\Pr(Z^* \le x) = \Pr(\tau \le G(x)) = G(x)$; thus Z^* has cumulative distribution function G, the generalized inverse of $\widetilde{q}$. For a nondecreasing left-continuous $\widetilde{q}$ the generalized inverse of its generalized inverse recovers the original function, so the quantile function of Z^* is exactly $\widetilde{q}(\cdot \mid s)$. $\square$

B Glossary

Written so that a reader from any one field can parse every other field's terms. Formulas referenced are in Paper A.

VaR / ES Value-at-Risk: the loss threshold exceeded with probability τ (a quantile with the sign flipped); Expected Shortfall: the average loss *given* VaR is exceeded (§2.1). FRTB capitalizes ES at $\tau = 2.5\%$.

Pinball (check) loss The asymmetric absolute loss (1) whose minimizer is the true quantile; the strictly proper score for quantile forecasts. Reported edges ("+2.7% of pinball") are percentage reductions in this loss.

GARCH-t / GJR / EWMA Parametric filters for conditional variance (Paper A); GJR adds a leverage term (down-moves raise vol more); EWMA is the fixed-decay special case (RiskMetrics); Student-t fixes the innovation shape.

HS / FHS Historical simulation: empirical quantiles of past returns. Filtered HS: empirical quantiles of GARCH-standardized residuals, rescaled by today's $\widehat{\sigma}_t$ — the regulatory workhorse; the special case of (Paper A) with no state dependence.

CAViaR Conditional autoregressive VaR: models the quantile itself as an autoregression, estimated on the pinball loss; the one non-nested rival.

EVT / GPD / POT Extreme-value theory: exceedances over a high threshold converge to the generalized Pareto distribution; fitting it (peaks-over-threshold) gives the tail extrapolation (Paper A).

Conformal prediction Distribution-free finite-sample correction sized on a held-out split (GRAFT-Q companion); guarantees coverage under exchangeability without any distributional assumption.

Quantile regression / IQN Estimating conditional quantiles by pinball ERM; the implicit quantile network (GRAFT-Q / Paper A) takes the level τ as an *input*, so one network carries the whole distribution and sampling is evaluation at random τ.

GBM (trees) Gradient-boosted regression trees on the pinball loss; the strongest point estimator on tabular features; cannot sample.

Amortization Training one model across many assets so inference for a new asset is evaluation, not estimation; the generative-Bayes reading is the inverse map $z \overset{D}{=} H(s, \tau)$ (Paper A).

Gibbs / generalized-Bayes posterior A posterior built from a loss instead of a likelihood, (2); ω tempers how fast the loss overwhelms the prior.

SBC Simulation-based calibration: simulate known truths, check the posterior's ranks are uniform; the audit in Algorithm 3.

Walk-forward, purge, embargo Refit on past data only, on a schedule; delete observations adjacent to the train/test boundary so overlapping windows cannot leak information.

DM test Diebold–Mariano: a t-test on the loss difference between two forecasters, robust to serial correlation.

MCS Model Confidence Set: the subset of models not statistically dominated at a given level — "co-best" means surviving in that set.

Kupiec / Christoffersen Regulatory exception tests: correct *number* of VaR breaches (unconditional coverage) and breaches arriving *independently* rather than in clusters.

DCC Dynamic conditional correlation: the standard multivariate GARCH benchmark for portfolio tails.

VRP; $\mathbb{P}$ vs. $\mathbb{Q}$ Variance risk premium: option-implied variance exceeds realized on average. $\mathbb{P}$ is the physical measure (what happens); $\mathbb{Q}$ the risk-neutral measure (what option prices imply); their wedge at the put wing is the crash premium studied in the companion paper.

Misspecification frontier The trailing residual-shape score (Paper A) and the empirical fact that the nonparametric edge concentrates in its top decile.